

IT314 – Lab 1

Assignment: Git & GitHub Fundamentals using a Bug Fix Workflow

Objective

The objective of this assignment is to gain hands-on experience with Git and GitHub by following a version control workflow. The emphasis of this lab is on correctly using Git and GitHub features.

Problem Statement

You have been provided with a C++ solution for the problem "Find First and Last Position of Element in Sorted Array." The solution intentionally contains bugs. Your task is to identify the issue, fix it, and manage the complete development process using Git and GitHub.

Assignment Requirements

1. Create a local Git repository containing the provided solution file.
2. Create a new GitHub repository and connect it to your local repository.
3. Push the initial version of the project to GitHub.
4. Identify the bugs in the provided solution.
5. Create a GitHub Issues describing the identified bugs.
6. Create a separate feature/fix branch to work on an issue.
7. Implement the fix in the solution file.
8. Commit your changes using a meaningful commit message.
9. Push the branch to GitHub.
10. Open a Pull Request from the feature branch to the main branch.
11. Merge the Pull Request after verifying the changes.
12. Ensure the final repository reflects a clean and meaningful Git history.

Files Provided

- first_last_occurrence.cpp (contains intentional bugs)

Submission

- GitHub repository link

- Screenshot of the GitHub Issues page
- Screenshot of the Pull Request page
- Screenshot of the final commit history
- Screenshot of the corrected program execution/output

Important Notes

The Git workflow has already been demonstrated during the lab session through the demo video. This assignment evaluates your ability to apply that workflow independently.